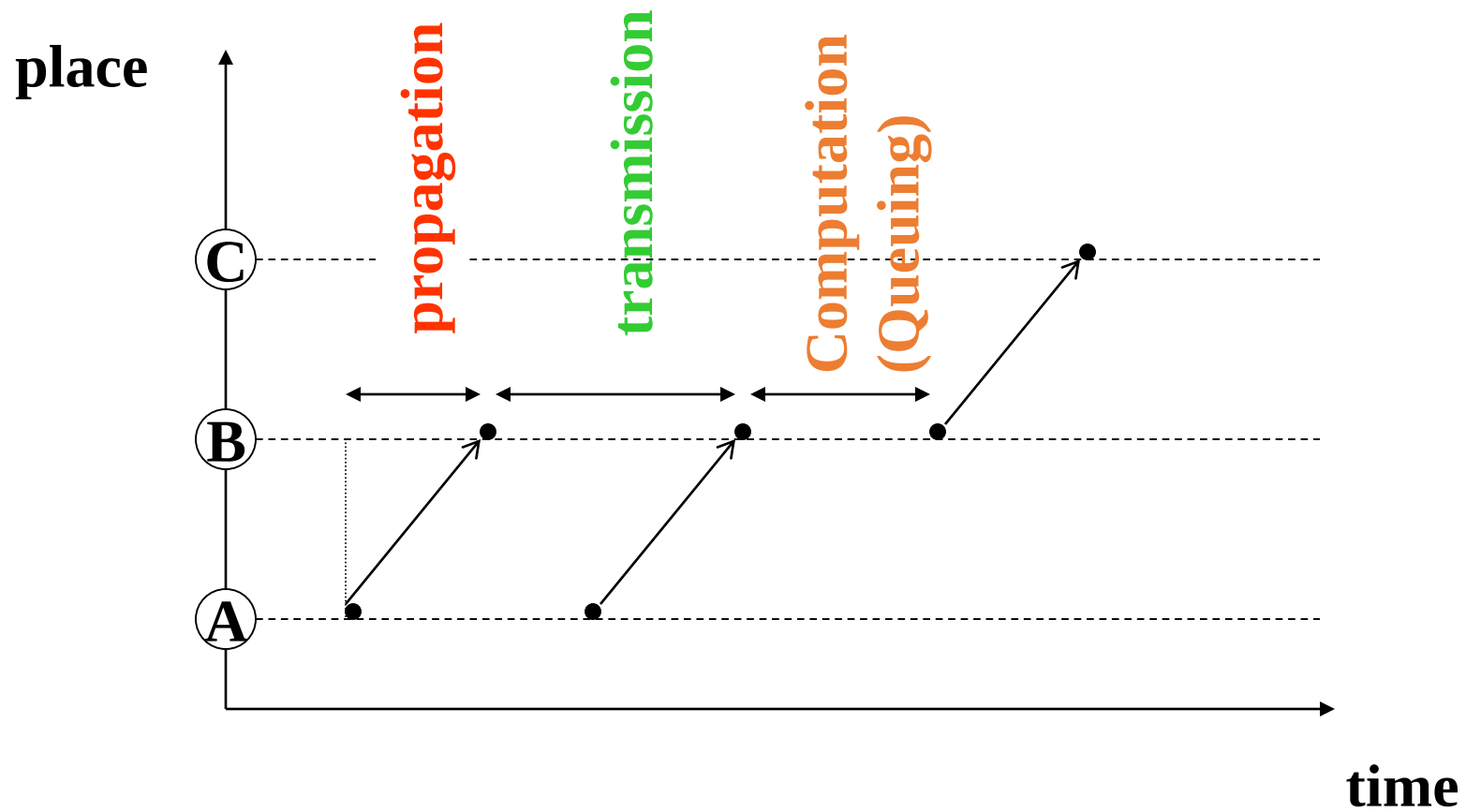


Queuing Theory

Introduction

- Many jobs share the system resources such as CPU, disks, etc, in a computer system.
- Queuing theory helps in determining the time that the jobs spend in various queues in the system.
- What is queuing theory?
- Mathematical analysis of queues and waiting times in stochastic systems.
 - Used extensively to analyze production and service processes exhibiting random variability in market demand (arrival times) and service times.
- Queues arise when the short term demand for service exceeds the capacity
 - Most often caused by random variation in service times and the times between customer arrivals.
 - If long term demand for service $>$ capacity the queue will explode!

Delay Models



Queue Model

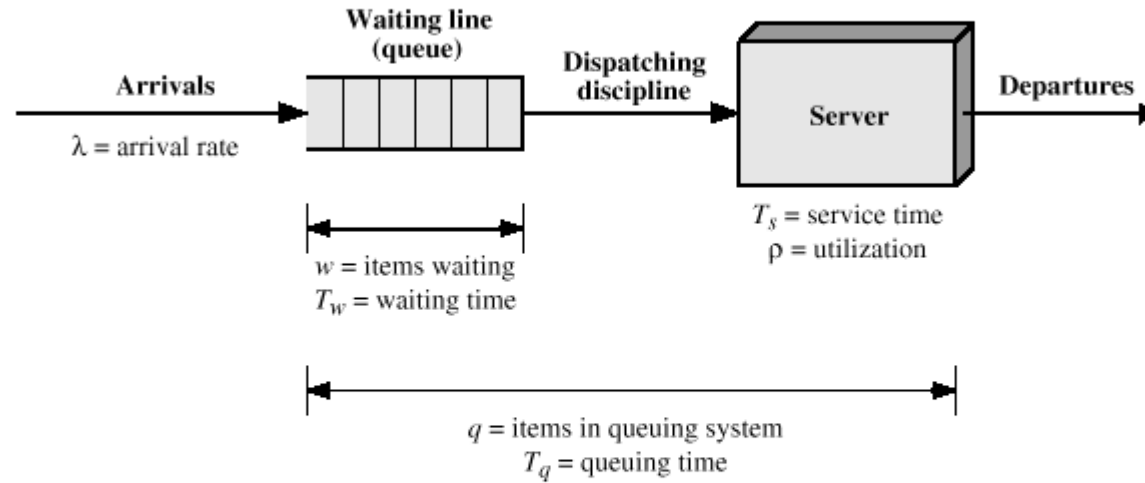


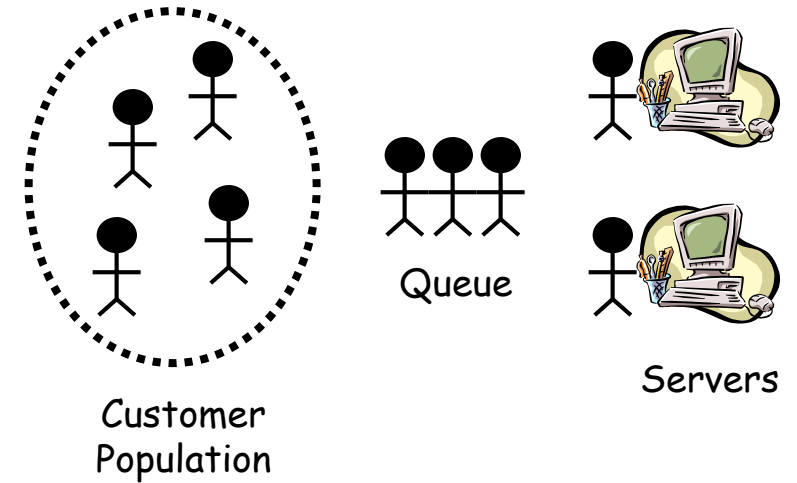
Figure 7.2 Queuing System Structure and Parameters for Single-Server Queue

Queuing Theory

- View network as collections of queues
 - FIFO data-structures
- Queuing theory provides probabilistic analysis of these queues
- Examples:
 - Average length
 - Probability queue is at a certain length
 - Probability a packet will be lost

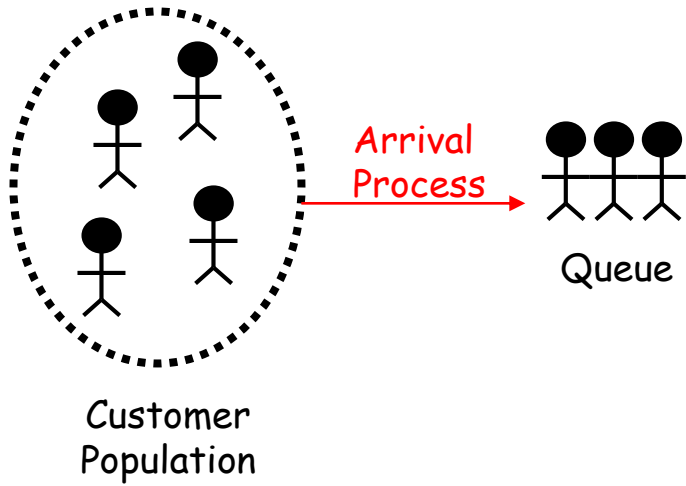
Queuing Notation

- For queuing analysis, need to specify:
 - Population size
 - Number of servers
 - System capacity
 - Arrival process
 - Service time distribution
 - Service discipline



Notation

- Population size
 - Potential customers who can enter
 - Most real systems finite but easier to analyze if infinite
- Number of servers
 - Can be one or more
 - Assume identical, but if not then separate queuing system for each
- System capacity
 - Number that can wait plus be served
 - The maximum number of customers who can stay in the system
 - Includes those waiting for service as well as those receiving service
 - Most systems have finite queue length, but easier to analyze if infinite



- Arrival process
 - Students arrive at $t_1, t_2, \dots, t_j$
 - Interarrival times are
$$\tau_j = t_j - t_{j-1}$$
 - Usually assume independent, identically distributed (IID)
 - Most common are Poisson arrivals
 - IID and exponentially distributed ($f(x) = \lambda e^{-\lambda x}$)

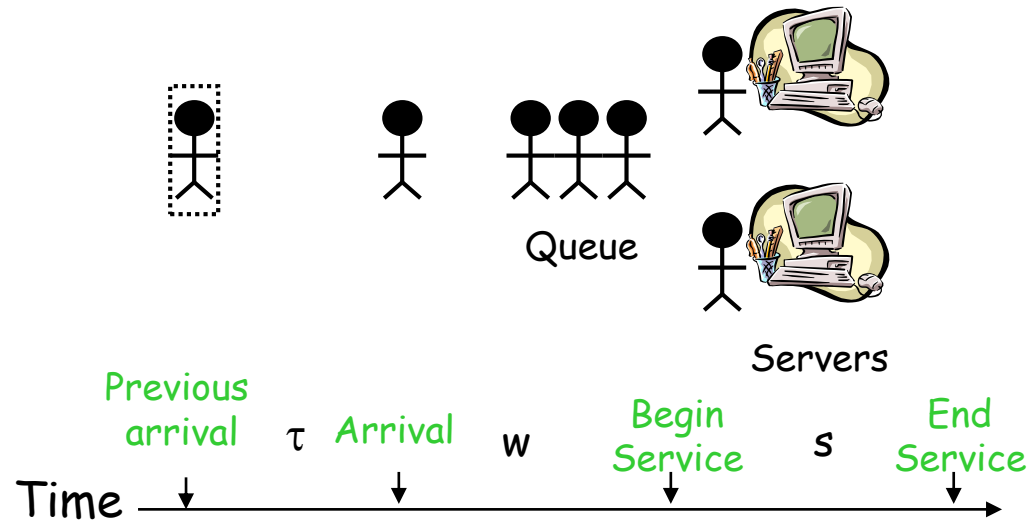
- Service time distribution
 - Amount of time each customer at server
 - Again, usually IID
 - Most common are exponential
- Service discipline
 - Order customers called for servicing
 - Most common is FCFS

Kendall notation

A/S/m/B/K/SD

- A is Arrival time distribution
- S is Service time distribution
- m is number of servers
- B is number of buffers
- K is population size
- SD is service discipline
- Some typical times used:
 - M Exponential
 - M means “memoryless” in that current arrival independent of past
 - D Deterministic
 - G General
 - Valid for all

Variables for all queues



- τ = interarrival time
- λ = mean arrival rate
 - $= 1/E[\tau]$
 - Can sometimes depend upon jobs in system
- s = service time per job
- μ = mean service rate per server
 - $= 1/E[s]$, total rate $m\mu$

n_q = number of jobs waiting in queue
 n_s = number of jobs receiving service
 n = number of jobs in system
 $n = n_q + n_s$
 r = response time
 w = waiting time

Note, all except μ and λ are random

Rules for All Queues (1 of 4)

- *Stability Condition*

- If the number of jobs becomes infinite, system unstable. For stability, mean arrival rate less than mean service rate

$$\lambda < m\mu$$

- Does not apply to finite queue or finite population systems
 - Finite population cannot have infinite queue
 - Finite queue drops if too many arrive so never has infinite queue

Rules for All Queues (2 of 4)

- *Number in System versus Number in Queue*

- Number of jobs is equal to waiting and servicing

$$n = n_q + n_s$$

- Also means:

$$E[n] = E[n_q] + E[n_s]$$

- So mean number of jobs is equal to mean number in queue plus mean number being serviced

$$\text{Var}[n] = \text{Var}[n_q] + \text{Var}[n_s]$$

- Variance of jobs equal to variance of queue + svc
- Also, service rate of servers independent of jobs in queue

$$\text{Cov}(n_q, n_s) = 0$$

Rules for All Queues (3 of 4)

- *Number versus Time*

- If jobs are not lost due to buffer overflow the mean jobs is related to response time as:

mean jobs in system = arrival rate x mean response time

- Similarly

mean jobs in queue = arrival rate x mean waiting time

- Above equations known as “**Little’s Law**”
- For finite buffers can use effective arrival rate (ignoring drops)

Rules for All Queues (4 of 4)

- *Time in System versus Time in Queue*

- Time spent in system is sum of queue and service time

$$r = w + s$$

- In particular:

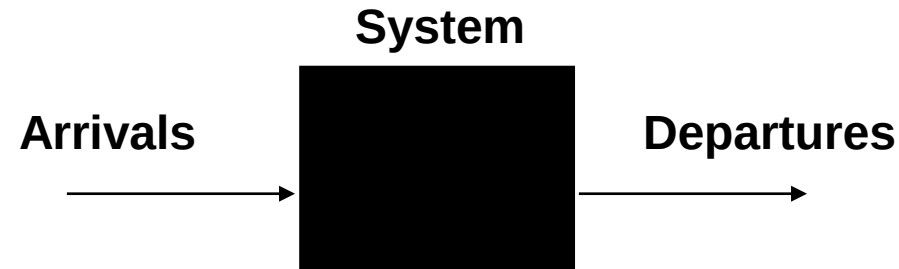
$$E[r] = E[w] + E[s]$$

- If service rate independent of jobs in queue

$$\text{Cov}(w,s) = 0$$

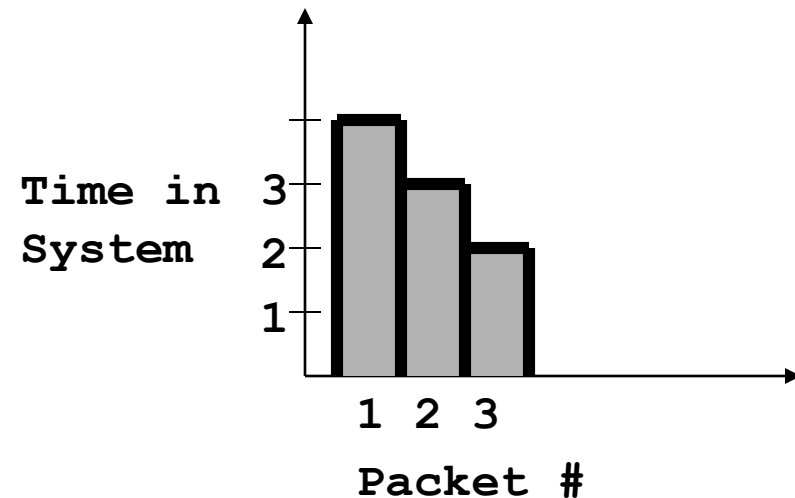
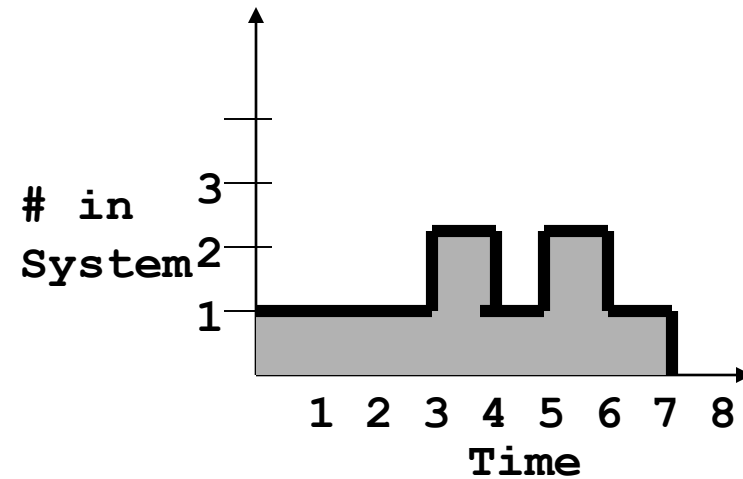
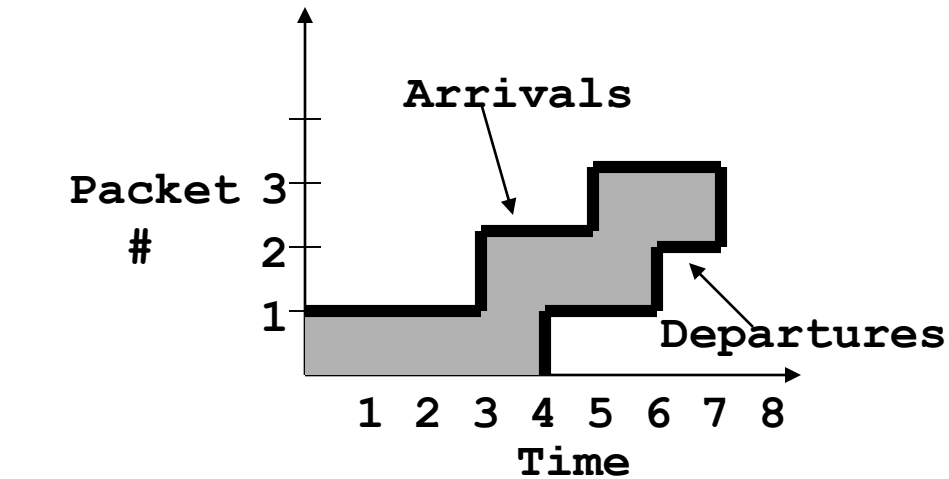
$$\text{Var}[r] = \text{Var}[w] + \text{Var}[s]$$

Little's Law



- Little's Law:
Mean number tasks in system = arrival rate x mean response time
 - Observed before, Little was first to prove
- Applies to any system in equilibrium, as long as nothing in black box is creating or destroying tasks
- Monitor the system for a time interval T and keep a log of arrival and departure times of each individual job

Proving Little's Law

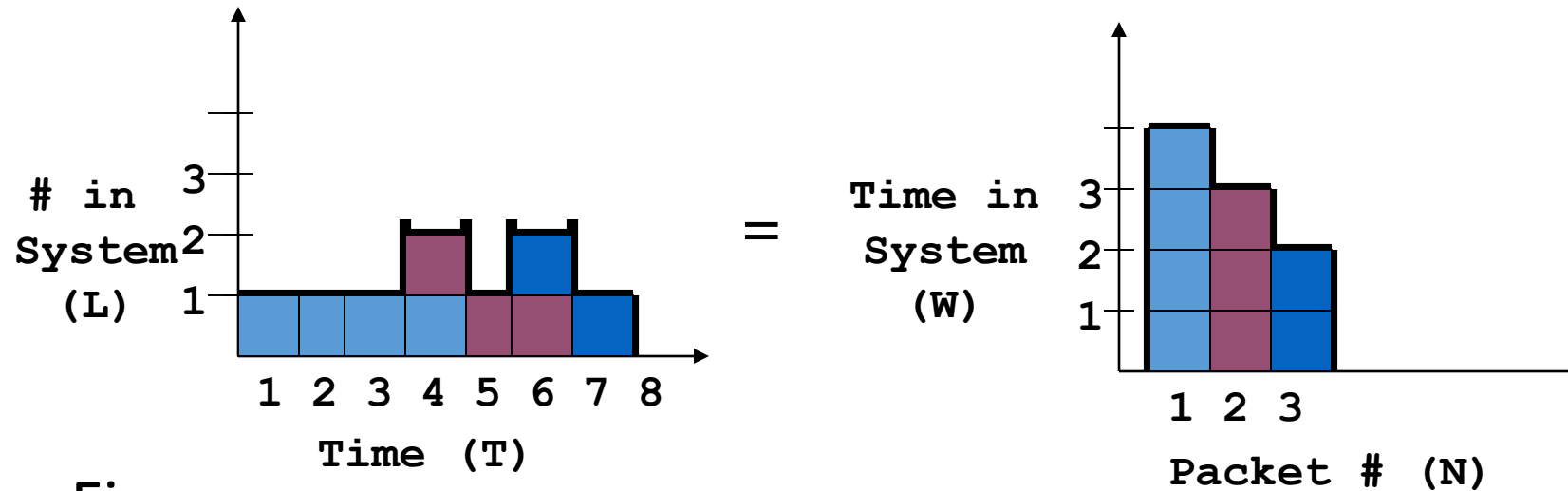


J = Shaded area = represent total time spent inside the system by all jobs. All three areas are equal

Definitions

- J: “Area” from previous slide
- N: Number of jobs (packets)
- T: Total time
- λ : Average arrival rate
 - N/T
- W: Average time job is in the system
 - $= J/N$
- L: Average number of jobs in the system
 - $= J/T$

Proof: Method 1: Definition



From Figure,

Mean time spent in the system = J/N ,

$$J = TL = NW$$

$$L = \left(\frac{N}{T}\right)W$$

$$L = (\lambda)W \quad \text{Little's Law}$$

What is a Stochastic Process?

- Stochastic Process: is a family of random variables $\{X(t) \mid t \in T\}$ (T is an index set; it may be discrete or continuous)
- Values assumed by $X(t)$ are called *states*.
- *State space* (I): set of all possible states
- Sometimes called a random process or a chance process

Types of Stochastic Processes

- ***Discrete-State and Continuous-State Process***
 - Depends upon values its state can take
 - If finite or countable $\rightarrow$ discrete
 - Ex: jobs in system $n(t)$ can only take values 0, 1, 2 ... countable, so discrete-state process
 - Also called a *stochastic chain*
 - Ex: waiting time $w(t)$ can take any real value, so continuous-state process
- ***Markov Process***
 - If future states depend only on the present and are independent of the past then called *markov process*
 - Makes it easier to analyze since do not need past trajectory, only present state
 - Also memory-less in that don't need length of time in current state

Types of Stochastic Processes

- ***Birth-Death Process***

- Markov in which transitions restricted to neighboring states only are called *birth-death process*
- Can represent states by integers, s.t. process in state n can only go to state $n+1$ or $n-1$
- Ex: jobs in queue with single server can be represented by birth-death process
 - Arrival (birth) causes state to change by $+1$ and departure after service (death) causes state to change by -1
 - Only if arrive individually, not in batch

Types of Stochastic Processes

- *Poisson Processes*

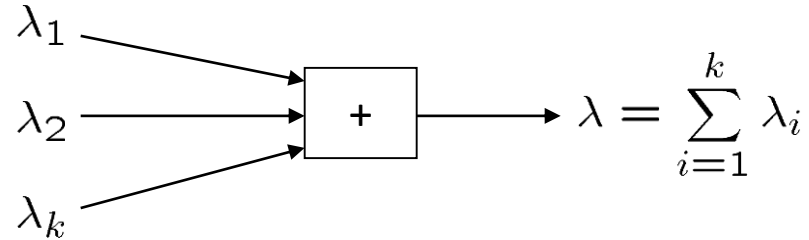
- If interarrival times are IID and exponentially distributed, then number of arrivals over interval $(t, t+x)$ has a Poisson distribution
→ *Poisson Process*
- Popular because arrivals are memoryless
- Also:
 - Merging k Poisson streams with mean rate λ_i gives another Poisson stream with mean rate:

$$\lambda = \sum \lambda_i$$

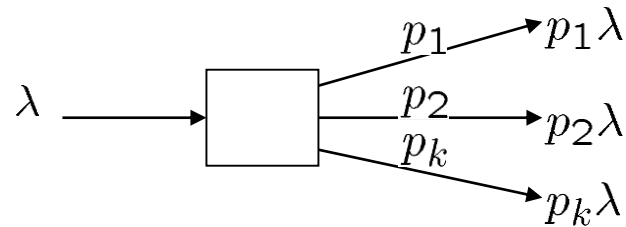
Types of Stochastic Processes

- *Poisson Processes* (continued)
 - Also
 - If Poisson stream split into k substreams with probability p_i , each substream is Poisson with mean rate λp_i
 - If arrivals to single server with exponential service times are Poisson with mean λ , departures are also Poisson with mean λ , if $(\lambda < \mu)$
 - Same relationship holds for m servers as long as total arrival rate less than total service rate

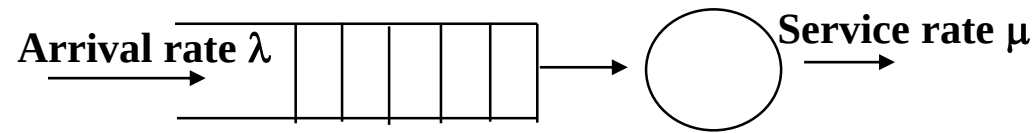
- Merging of k Poisson streams with mean rate λ_i results in a Poisson stream with mean rate λ given by



- Splitting of a Poisson stream into k substreams such that the probability of a job going to the i^{th} substream is P_i , each substream is also Poisson with a mean rate of $P_i\lambda$



- If the arrivals to a single server with exponential service time are Poisson with mean rate λ , the departures are also Poisson with the same rate λ , provided that $\lambda < \mu$

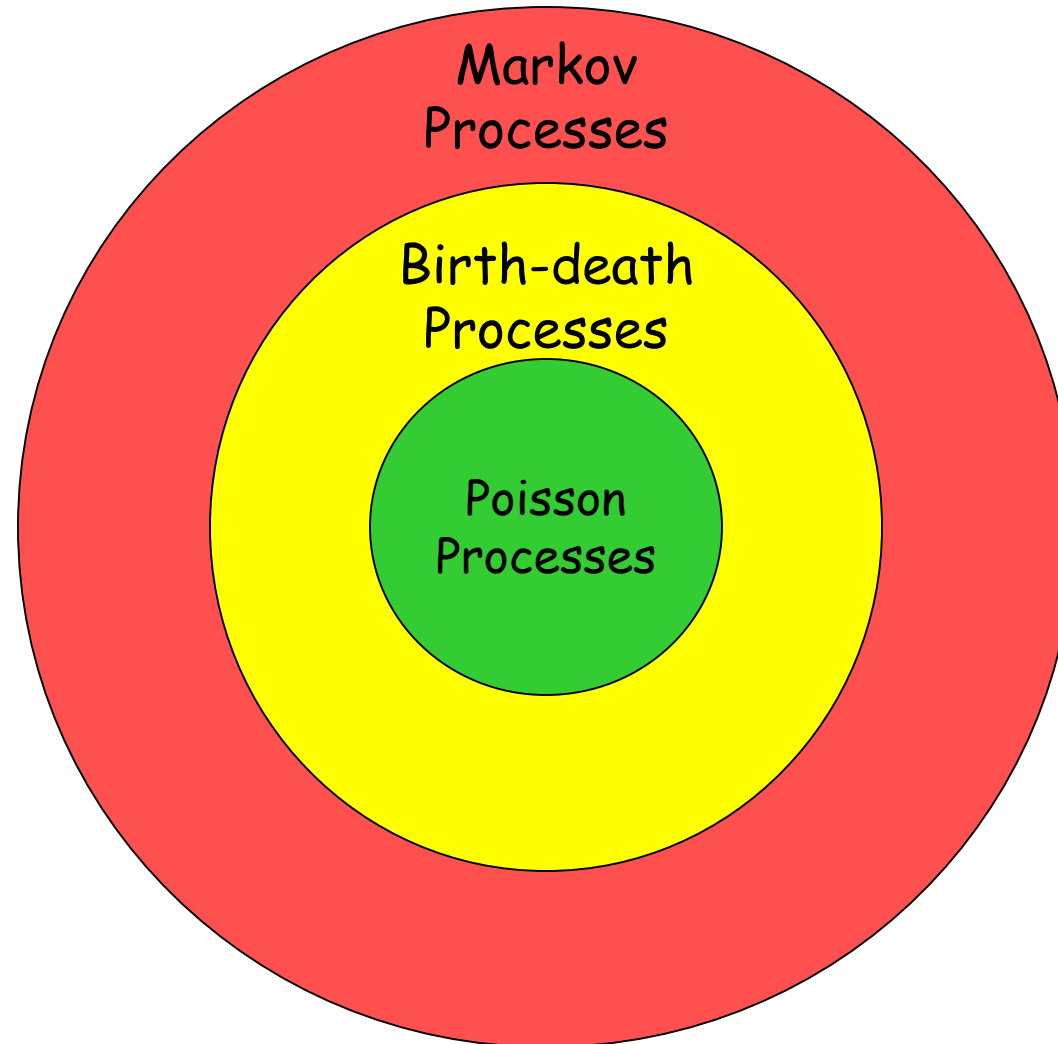


- If the arrivals to a service facility with m service centers are Poisson with a mean rate λ , the departures also constitute a Poisson stream with same rate λ , provided $\lambda < \sum \mu_i$

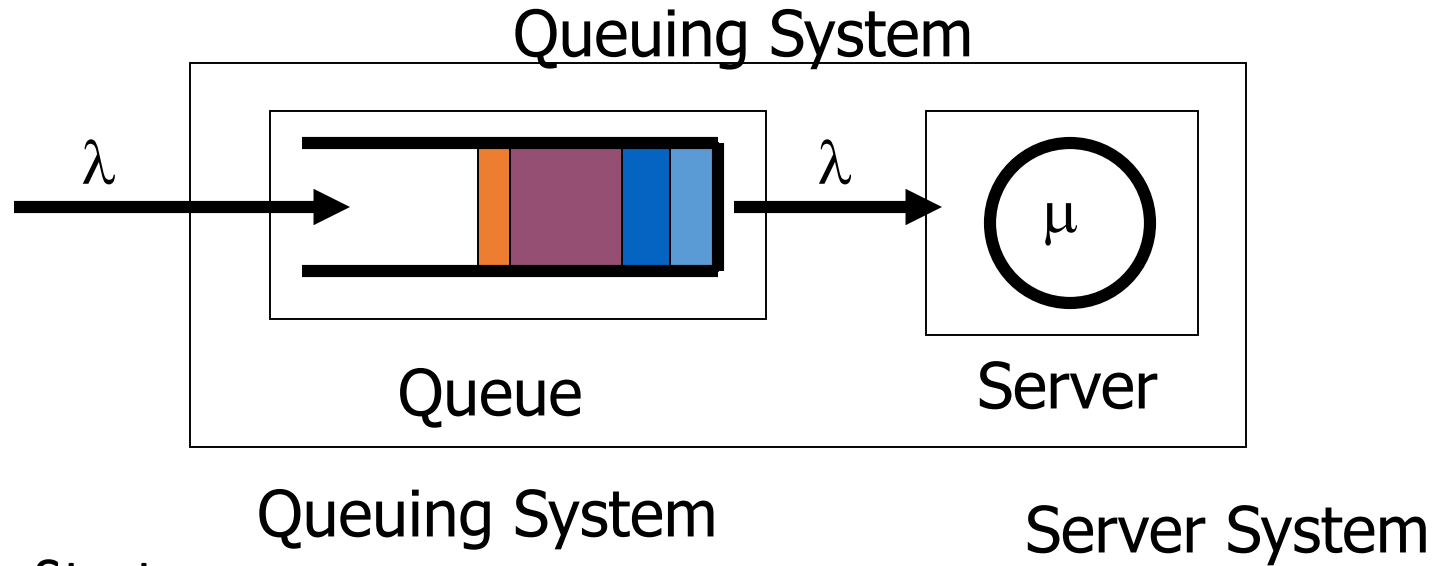
PASTA property

- PASTA: Poisson Arrivals See Time Averages
- It refers to the expected state of a queuing system as seen by an arrival from a Poisson process
- Meaning: When a customer arrives, it finds the same situation in the queueing system as an outside observer looking at the system at an arbitrary point in time.
- If an external observer samples a system at a random instant:
- $P(\text{system state} = x) = P(\text{state as seen by a Poisson arrival is } x)$
- Therefore, the expected value of any parameter of the queue at the instant of a Poisson arrival is simply the long-run average value of that parameter

Types of Stochastic Processes



Model Queuing System



Strategy:

Use Little's law on both the complete system and its parts to reason about average time in the queue

Kendall Notation

- Six parameters in shorthand
 - First three typically used, unless specified
- 1. Arrival Distribution
 - Probability of a new packet arrives in time t
- 2. Service Distribution
 - Probability distribution packet is serviced in time t
- 3. Number of servers
- 4. Total Capacity (infinite if not specified)
- 5. Population Size (infinite)
- 6. Service Discipline (FCFS/FIFO)

Distributions

- M: Exponential
 - D: Deterministic (e.g. fixed constant)
 - E_k : Erlang with parameter k
 - H_k : Hyperexponential with param. k
 - G: General (anything)
-
- M/M/1 is the simplest 'realistic' queue

Queueing Model Classification

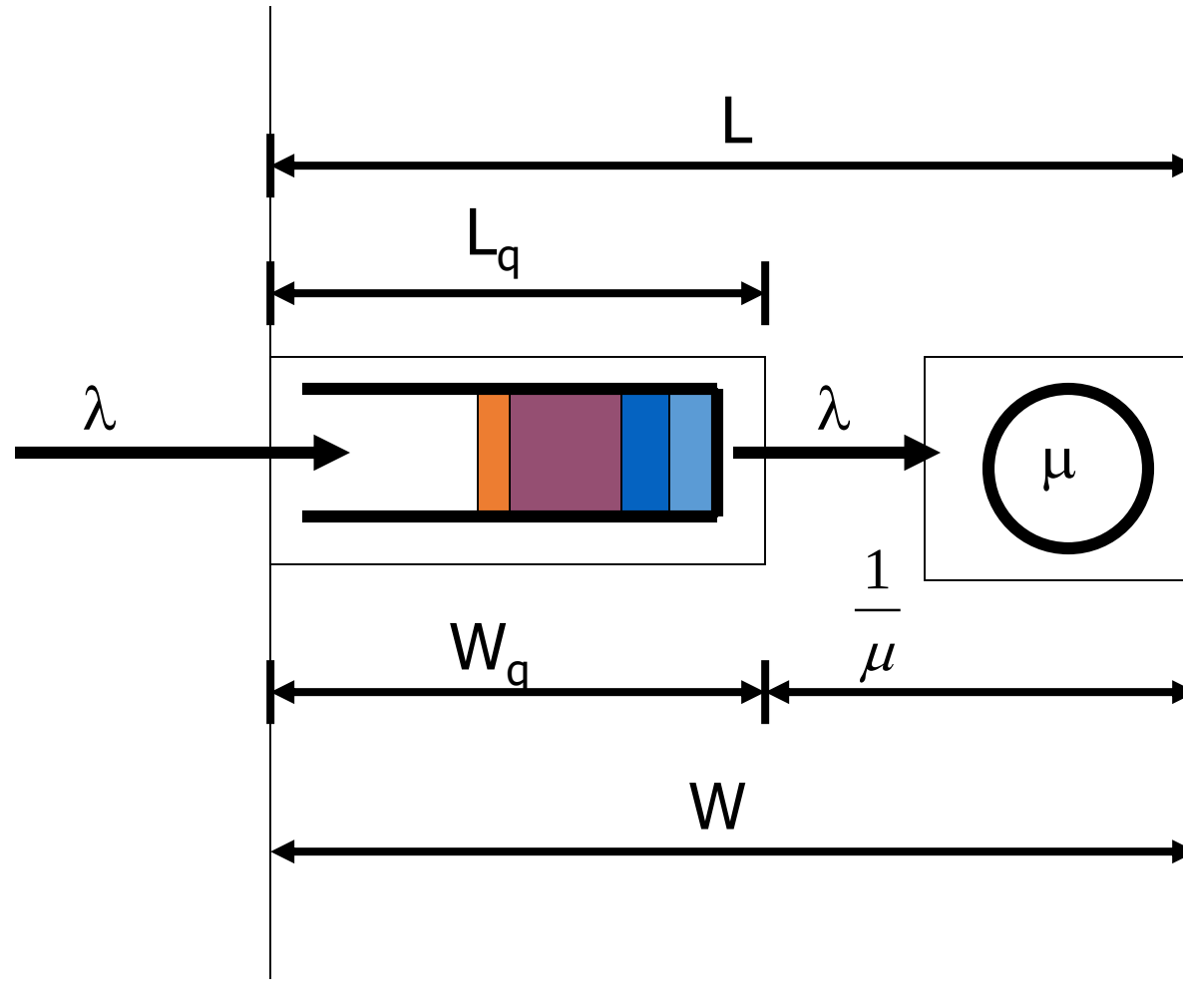
<i>Arrival Process / Service Time / Servers / Max Occupancy</i>			
Interarrival times τ	Service times X	1 server	K customers
M = exponential	M = exponential	c servers	unspecified if
D = deterministic	D = deterministic	infinite	unlimited
G = general	G = general		
Arrival Rate:	Service Rate:		
$\lambda = 1 / E[\tau]$	$\mu = 1 / E[X]$		

Multiplexer Models: M/M/1/K, M/M/1, M/G/1, M/D/1

Trunking Models: M/M/c/c, M/G/c/c

User Activity: M/M/ ∞ , M/G/ ∞

M/M/1 queue model



The M/M/1 System

